

NAME

mbm_trnplot – Macro to plot map views of an **mbtrnpp** terrain relative navigation (TRN) session.

VERSION

Version 5.0

SYNOPSIS

mbm_trnplot [-i -ggridfile -jcrs --mission=missionname -d -h]

DESCRIPTION

mbm_trnplot generates map-view plots comparing an **mbtrnpp** terrain relative navigation (TRN) session's estimated vehicle track against reference bathymetry, using **mbm_grdplot**, **GMT**, and (for a second, locally regridded comparison) **mbgrid**.

The macro searches an input session directory (**-i/--input**; by default the directory pointed to by a symbolic link named "mbtrnpp-latest" in the current directory if one exists, otherwise the current directory) for **mbtrnpp** session log files matching the pattern "*_trn.txt". For each session log found, the macro determines a reference topography grid to plot the track over. This grid can be given explicitly with **-g/--grid**; otherwise the macro looks for a symbolic link named "mbtrnpp-grid" in the current directory, and failing that it parses the session log's own header comments for a line beginning "## Reference topography model:" to locate the TRN octree/grid used during the session (translating a ".bo" octree reference to the corresponding "source_grid.grd" grid, or otherwise appending "/source_grid.grd" to a bare tileset directory name).

Each session log's data lines are parsed to extract, per ping: the navigation-only position, the TRN-estimated position, and lon/lat equivalents of both when present. TRN offset estimates flagged in the log as "CNV", "USE", or "RNT" are additionally written out as separate subsets so that converged, actively-used, and reinitialized fixes can be distinguished on the plot. These extracted positions are written to a set of companion text files in the input directory (with suffixes such as "_trn_Nxy.txt", "_trn_Txy.txt", "_trn_TxyC.txt", "_trn_TxyU.txt", "_trn_TxyR.txt", and lon/lat equivalents ending in "Nll"/"Tll"/etc.), and the overall bounding box of the plotted positions (padded by 10%) is computed.

mbm_trnplot then constructs and runs an **mbm_grdplot** command to generate a shaded-relief map of the reference topography with the navigation-only track, the raw TRN-estimated track, and the CNV/USE/RNT position subsets overlaid, converts the resulting PostScript plot to a JPEG image with **gmt psconvert**, and (unless **-d/--display** is given, in which case the plot is also displayed immediately) runs the generated plotting script in non-interactive mode. The macro then repeats this process a second time against a bathymetry grid built on the fly from any MB1-format (".mb1") processed data files found in the input directory, using **mbm_makedatalist**, **mbdatalist**, and **mbgrid** (with the projection set by **-j/--crs**), so that TRN performance can also be compared against bathymetry gridded directly from the vehicle's own logged soundings.

The plot title uses the mission name given with **--mission** if supplied, otherwise a session name derived from the log filename.

MB-SYSTEM AUTHORSHIP

David W. Caress
 Monterey Bay Aquarium Research Institute
 Dale N. Chayes
 Center for Coastal and Ocean Mapping
 University of New Hampshire
 Christian do Santos Ferreira
 MARUM - Center for Marine Environmental Sciences

University of Bremen

OPTIONS

-h, --help

This "help" flag causes the macro to print out a description of its command line options and then exit immediately.

-d, --display

Causes each generated plot to be displayed interactively as soon as it is produced. If this option is not given, the plotting scripts are instead run in non-interactive (batch) mode.

-iinputdirectory, --input=inputdirectory

Sets the path to the directory containing the **mbtrnpp** session log(s) and associated processed results to be plotted. If this option is not given (or is given the literal value "mbtrnpp-latest"), the macro uses the directory pointed to by a symbolic link named "mbtrnpp-latest" in the current directory if one exists, and otherwise uses the current directory.

-ggridfile, --grid=gridfile

Sets the path to the reference topography grid plotted beneath the TRN track. If this option is not given (or is given the literal value "mbtrnpp-grid"), the macro looks for a symbolic link named "mbtrnpp-grid" in the current directory, and failing that attempts to determine the reference grid from the "## Reference topography model:" header comment recorded in the session log.

-jcrs, --crs=crs

Sets the coordinate reference system code passed to **mbgrid**'s **-J** option when regridding the session's MB1-format processed data for the second comparison plot. Default: *crs* = "U" (an automatically selected UTM projection).

--mission=missionname

Sets a mission name used in the plot titles in place of the session name derived from the log filename. Note: in the installed version of this macro the short option **-m** does not itself take an attached value (a current limitation of its command line option parsing); use the long form **--mission=missionname** to set this value.

SEE ALSO

mbsystem(1), **mbtrnpp(1)**, **mbm_grdplot(1)**, **mbgrdviz(1)**, **mbgrid(1)**

BUGS

Maybe. Maybe not.